

CatGen

Comparing DCGAN, AAE, and VQ-VAE on cat image generation

Your Name · Deep Learning Project III

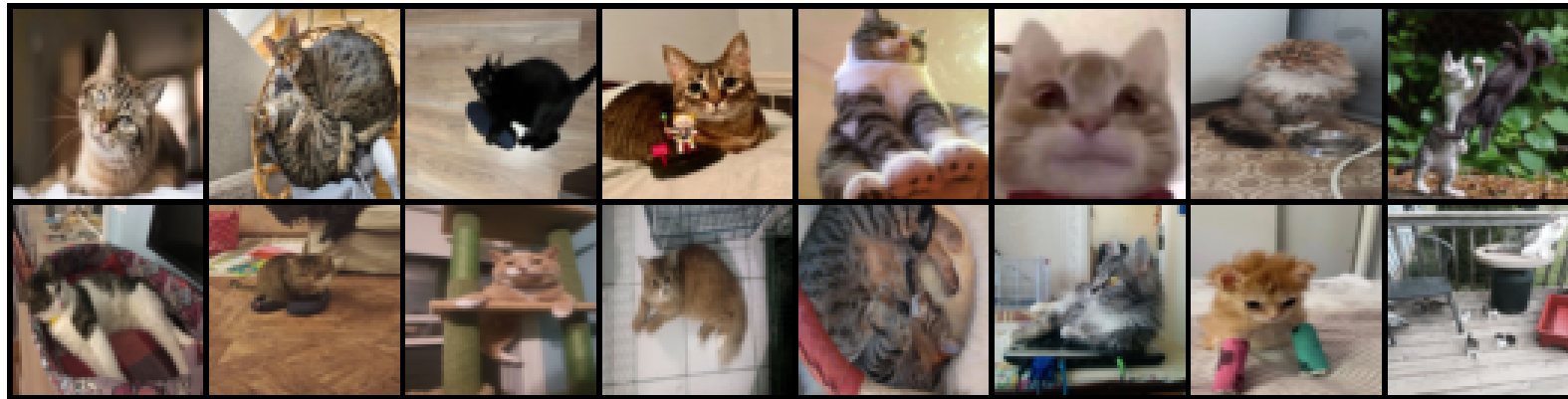
Goal

Question: Which lightweight generative model produces the best cat images under a fixed GPU budget?

- Three families: **DCGAN**, **Adversarial Autoencoder (AAE)**, **VQ-VAE**
- Metrics: **FID** + sample grids + latent interpolation (DCGAN/AAE)
- Extension: **cats + dogs** training for the **best** model only (128×128 phase)

Dataset

- **Cats:** Kaggle Cat Dataset (~10k images)
- **Dogs:** Dogs vs Cats (extension only)
- **Train / FID ref:** 3000 / 1000 cats (128×128); 1500 / 500 for 64×64 sweep on cluster
- **Preprocessing:** resize or crop → RGB → normalize to $[-1, 1]$; horizontal flip (train)



Training cats after preprocessing (use Kaggle preview after cluster rsync)

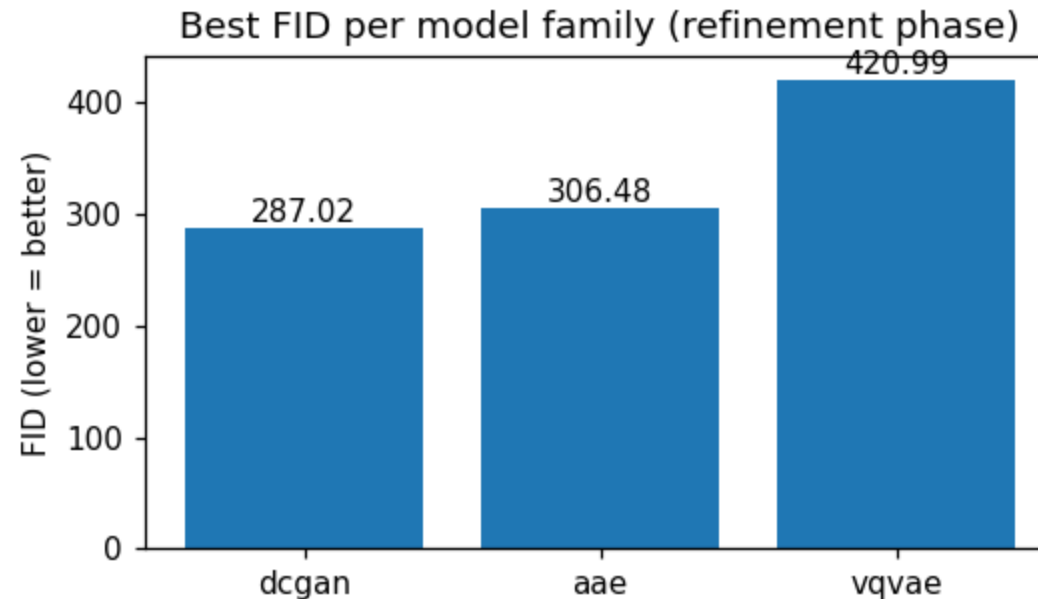
Models (64×128 capable)

Model	Idea	Latent	Params (order)
DCGAN	Generator + discriminator	Continuous z	~9.6M
AAE	Conv AE + adversarial latent D	Continuous z	~7.7M
VQ-VAE	Encoder → codebook → decoder	Discrete codes	~0.5M

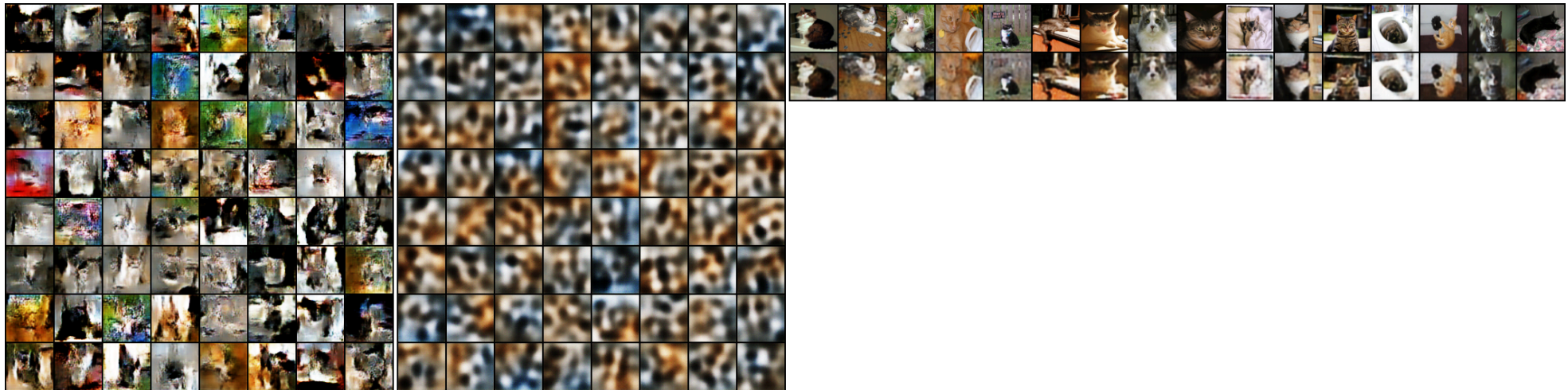
Same stack: PyTorch Lightning, reproducible `run_id`, early stopping on AAE/VQ (`recon`).

Part A — What we tried (64×64)

- **Sweep:** ~10 configs per family (z, lr, codebook K) on **1500** train cats
- **Refine:** 80 epochs, DCGAN **upsample+conv** G, label smoothing, separate G/D LR



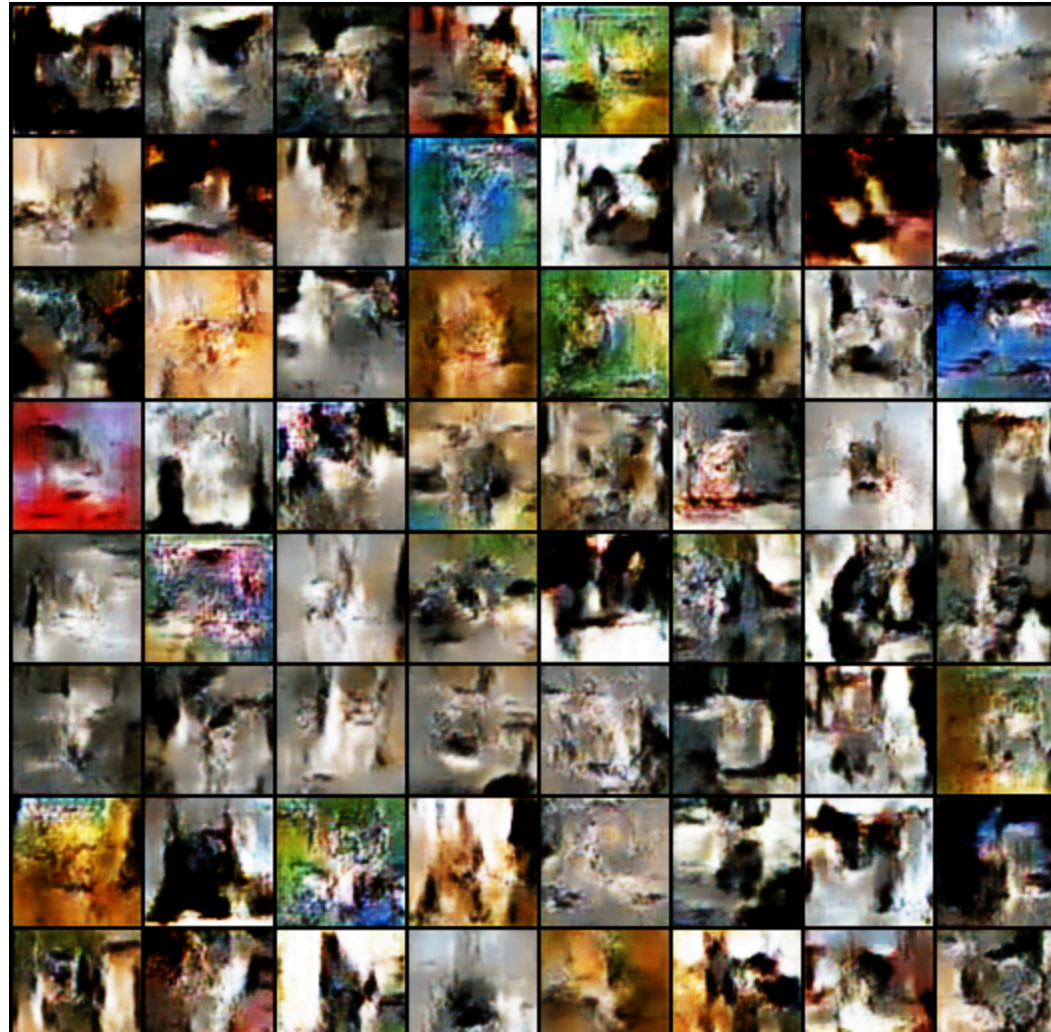
Part A — Sample grids (64×64, eval)



DCGAN · AAE · VQ-VAE (left → right)

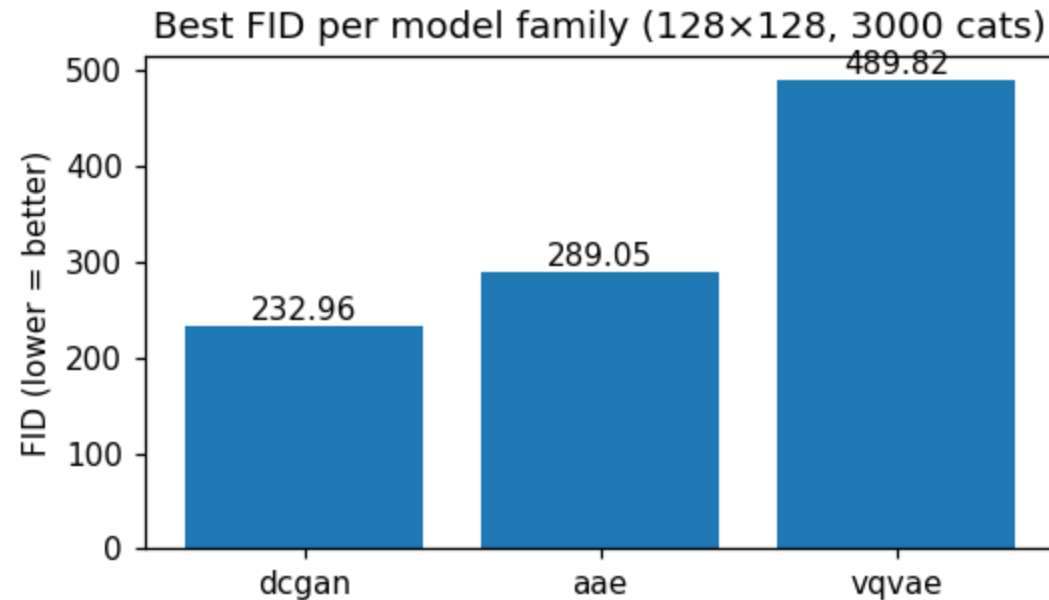
Part A — Outcome

- Best cats-only FID still ~275–420 (DCGAN lowest)
- **Refinement did not fix visuals** — samples remain blob-like textures

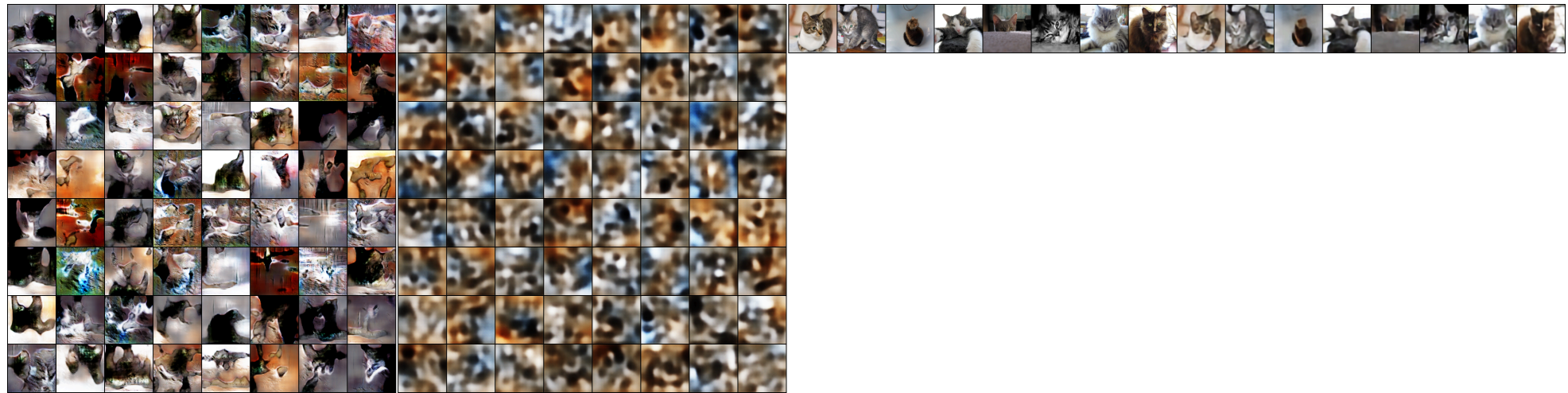


Part B — 128×128, 3000 cats

- Retrain all three at 128×128 on 3000 train cats (RandomResizedCrop aug)
- FID vs 1000 reference cats



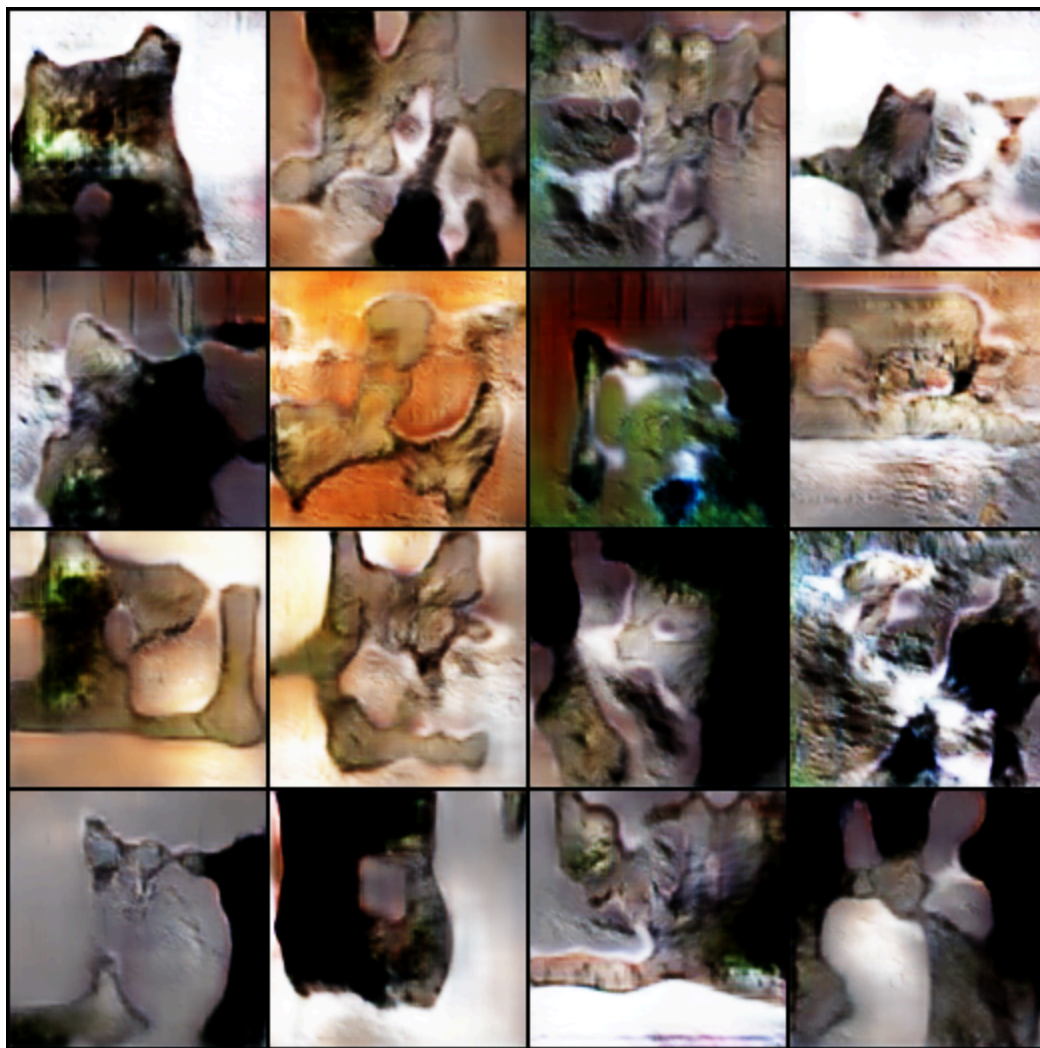
Part B — Sample grids (128×128, eval)



DCGAN · AAE · VQ-VAE (left → right)

Part B — DCGAN @ 128×128

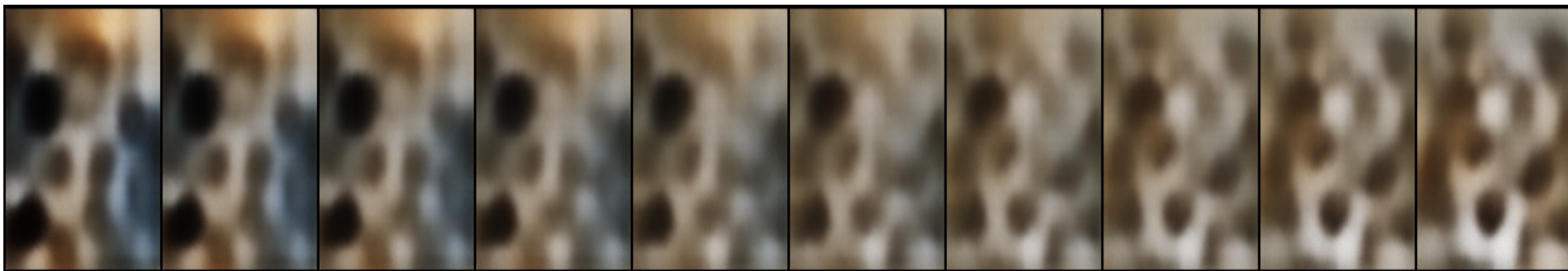
`dcgan_e9574605_42` — FID ~233 (best so far)



Part B — Latent interpolation



DCGAN @ 128×128



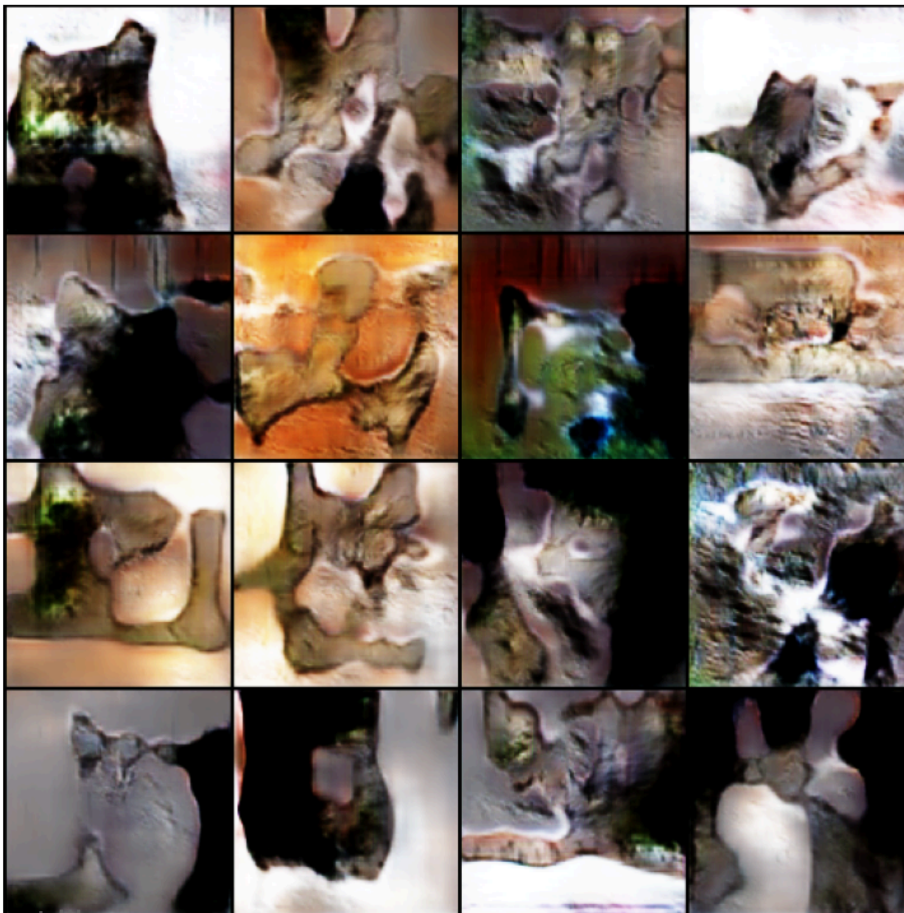
AAE @ 128×128

Part B — Cats vs dogs (best model)

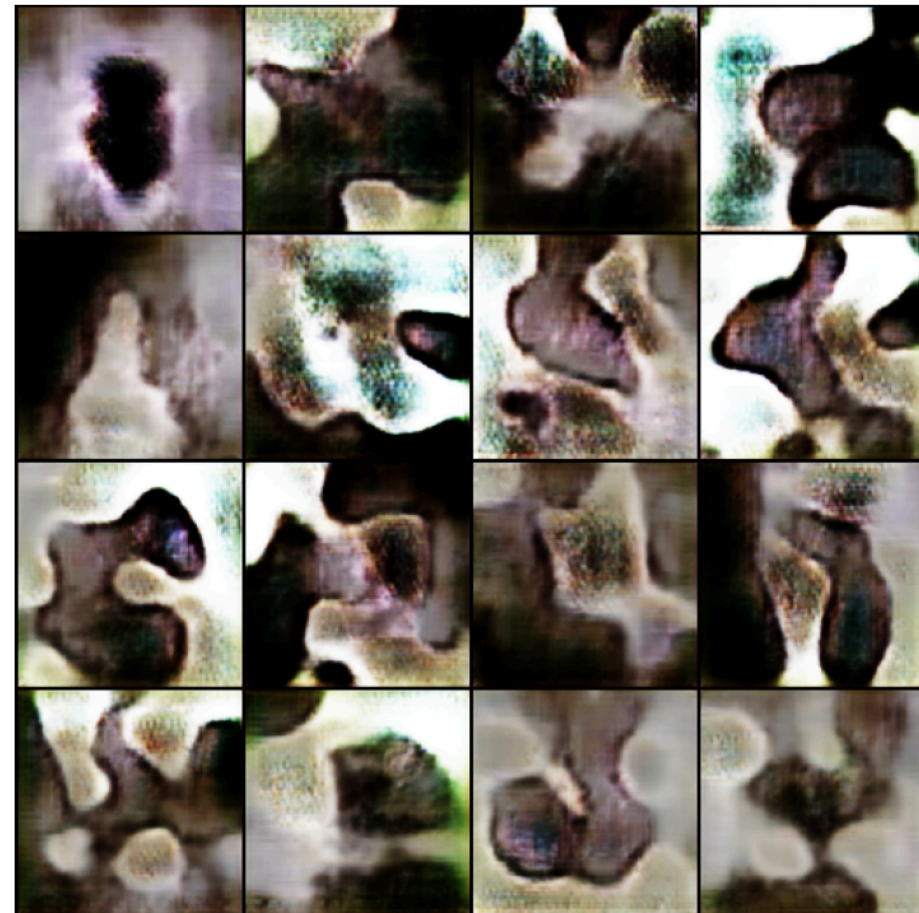
Same architecture; cats-only vs mixed training (DCGAN extension)

Extension: same architecture, different training data — 128×128, best of three families

Cats only (DCGAN)
FID=233.0



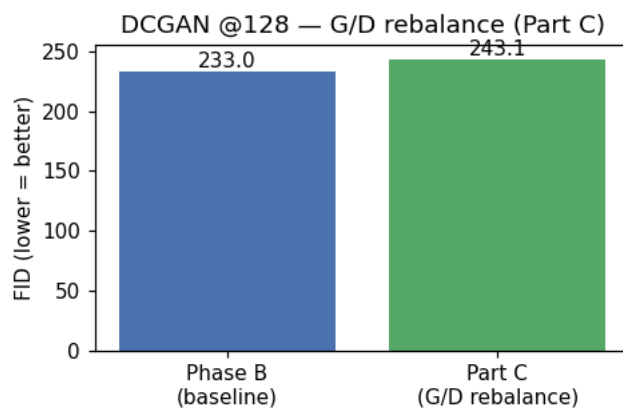
Cats + dogs (DCGAN)
FID=271.0



Part C — G/D rebalance @128

Phase B DCGAN: `loss_g` plateau, discriminator winning → try rebalancing G vs D (same 3k cats, 80 epochs)

	Phase B	Part C
<code>lr</code> (G)	1e-4	2e-4
<code>lr_d</code>	2e-4	1e-4
<code>label_smooth</code>	0.1	0.05

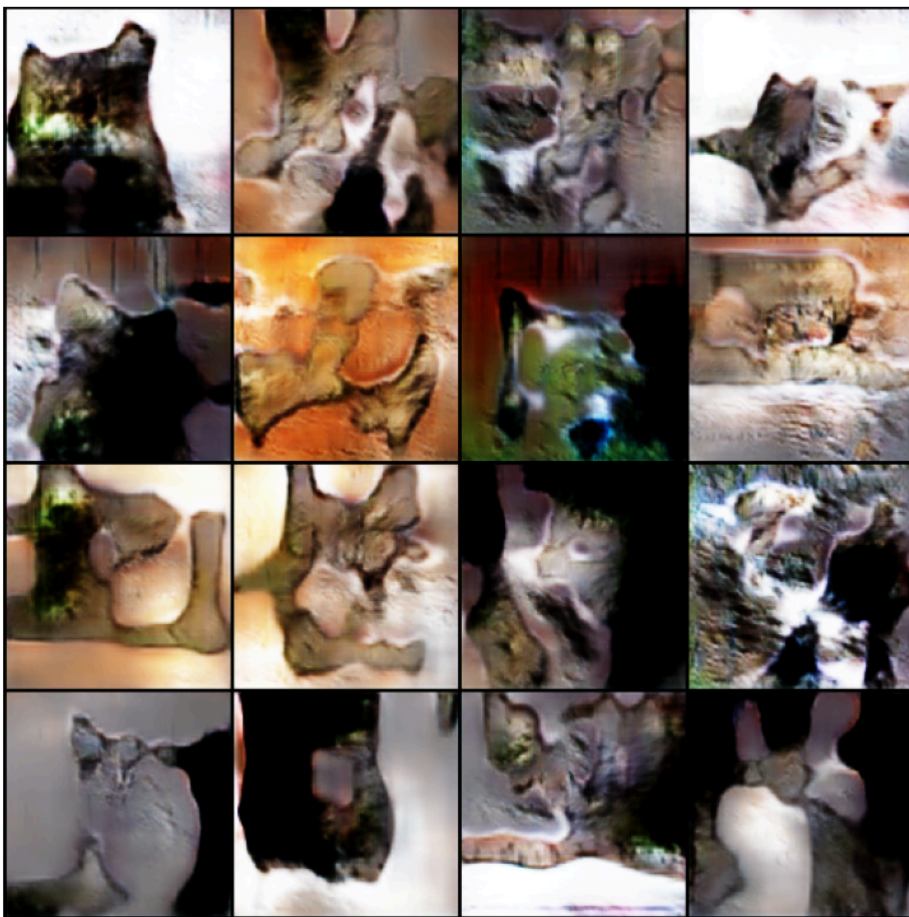


FID: 233 → 243 — rebalance did **not** improve on Phase B

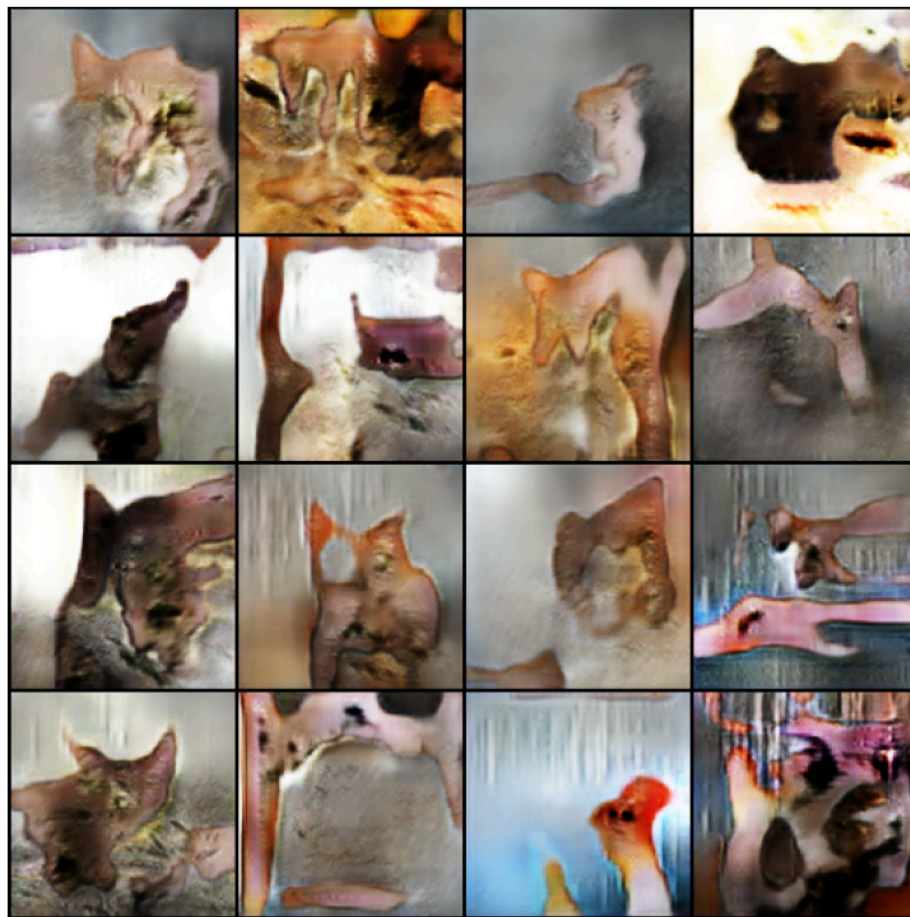
Part C — Baseline vs rebalance (eval samples)

DCGAN eval samples — baseline vs G/D rebalance

Phase B
FID=233.0



Part C
FID=243.1



Conclusions

Phase	Best model	Best cats FID
64×64 refine	DCGAN	~287
128×128	DCGAN	~ 233
128×128 G/D rebalance	DCGAN	~243 (worse)

1. **DCGAN** wins across phases (AAE/VQ-VAE higher FID)
2. **64×64 sweep + refine** = fair comparison, **not** good visuals
3. **128×128 + more data** = clear improvement
4. **G/D rebalance** did not beat Phase B — simple LR/smoothing tweak insufficient

Limitations

- Small **custom** nets — not StyleGAN / diffusion
- **FID not comparable** across 64 vs 128 (different resolution & splits)
- **FID-only** — no human study; **VQ-VAE** has no latent interpolation in our pipeline
- Cats-vs-dogs extension only for **best** model at 128; cluster used smaller splits in Part A

Thank you

Questions?

github.com/AdamKaniasty/DeepLearningCatGen